

CS1540 Algorithmic design

Steven Chaplick, David Mestel

Homework 4.

1 Introduction

Sorry, no Codegrade instance this time!

2 n -doku

An n -doku requires you to fill in an n^2 -by- n^2 grid (built out of and n -by- n grid of n -by- n small grids) with the numbers 1 to n such that each row, each column and each of the n^2 small grids contains each number at most once. So the classic Sudoku puzzle is a 3-doku.

1. Design a backtracking algorithm to solve n -doku puzzles (i.e. given a partially filled grid find a solution if one exists or return IMPOSSIBLE if not).
2. (optional) Implement your algorithm in Java. How large puzzles can it solve?
3. Now for partially filled grids where the answer is IMPOSSIBLE, we are interested in maximum-size solutions, i.e. given a partially-filled grid we want to write in as many numbers as possible while following the n -doku rules. Formulate this problem as an ILP (hint: use variables $x_{i,j,k}$ with $1 \leq i, j \leq n^2$ and $1 \leq k \leq n$ where $x_{i,j,k} = 1$ if we write k in cell (i, j)).
4. What is the fractional relaxation of this ILP? Write pseudocode for a branch-and-bound algorithm using the fractional relaxation.
5. (very optional) Implement your algorithm in Java, using a suitable library for LP solving.